

Cálculos torque-motor para caminata

1. Lados triángulos naranjos (fijos)

```
In[ ]:= a1 = 90;  
        k1 = 295;  
        a2 = 50;  
        k2 = 50;
```

2. Lados triangulo principal

```
In[ ]:= bmin = 275;  
        bmax = 385;  
        k = Sqrt[k1^2 + k2^2] // N  
           |raíz cuadrada      |va  
        a = Sqrt[a1^2 + a2^2] // N  
           |raíz cuadrada      |v:
```

```
Out[ ]:= 299.207
```

```
Out[ ]:= 102.956
```

3. Ángulos triángulos naranjos (fijos)

```
In[ ]:= sola = Solve[a2^2 == a1^2 + a^2 - 2 * a1 * a * Cos[x] && 0 < x < Pi, x] // N  
           |resuelve                                |coseno      |número pi |v:  
        a3 = (x /. First[sola]) / Degree // N  
           |primero      |grado      |valor numérico
```

 **Solve:** Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

```
Out[ ]:= { {x -> 0.507099} }
```

```
Out[ ]:= 29.0546
```

```

In[ ]:= solk = Solve[k2^2 == k1^2 + k^2 - 2 * k1 * k * Cos[x] && 0 < x < Pi, x];
           [resuelve]                [coseno]                [número pi]
k3 = (x /. First[solk]) / Degree // N
           [primero]                [grado]                [valor numérico]

```

... Solve: Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

Out[]:= 9.61973

4. Ángulo rodilla máximo con largo husillo mínimo (no usar)

```

In[ ]:= sol = Solve[bmin^2 == a^2 + k^2 - 2 * a * k * Cos[x] && 0 < x < Pi, x] // N;
           [resuelve]                [coseno]                [número pi] [val]
beta = (x /. First[sol]) / Degree // N
           [primero]                [grado]                [valor numérico]

```

... Solve: Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

Out[]:= 66.5681

```

In[ ]:= alfa = 180 - beta - a3 - k3

```

Out[]:= 74.7576

5. Ángulo rodilla mínimo con largo husillo máximo

```

In[ ]:= sol = Solve[bmax^2 == a^2 + k^2 - 2 * a * k * Cos[x] && 0 < x < Pi, x] // N;
           [resuelve]                [coseno]                [número pi] [val]
beta = (x /. First[sol]) / Degree // N
           [primero]                [grado]                [valor numérico]

```

... Solve: Solve was unable to solve the system with inexact coefficients. The answer was obtained by solving a corresponding exact system and numericizing the result.

Out[]:= 141.326

```

In[ ]:= alfa = 180 - beta - a3 - k3

```

Out[]:= -4.9738×10^{-14}

6. Largo husillo con ángulo rodilla literatura (desde acá probar ángulos)

```

In[ ]:= alfa = 1;

```

```

In[ ]:= beta = -alfa + 180 - a3 - k3

```

Out[]:= 140.326

```
In[ ]:= sol = Solve[x^2 == a^2 + k^2 - 2 * a * k * Cos[beta °] && x > 0, x] // N;
           |resuelve          |coseno          |val
           |
           b = (x /. First[sol]) // N
           |primero          |valor numérico
Out[ ]:= 384.117
```

7. Vectores posición

```
In[ ]:= K = {0, 0, 0};
A = {-a2, a1, 0};
B = {-k2 * Cos[alfa °] - k1 * Sin[alfa °], k2 * Sin[alfa °] - k1 * Cos[alfa °], 0} // N
           |coseno          |seno          |seno          |coseno          |v
Out[ ]:= {-55.1408, -294.082, 0.}

In[ ]:= KA = A - K
KB = B - K
AB = B - A

Out[ ]:= {-50, 90, 0}

Out[ ]:= {-55.1408, -294.082, 0.}

Out[ ]:= {-5.14084, -384.082, 0.}
```

8. Brazo palanca (P = peso, M = momento extensor, r = brazo, F = fuerza)

```
In[ ]:= c = 0.49;
P = 90.36;

In[ ]:= M = c * P
r = Norm[Cross[KA, KB]] / Norm[AB] / 1000 // N
           |norma |producto vectorial |norma          |v
F = M / r

Out[ ]:= 44.2764

Out[ ]:= 0.0512

Out[ ]:= 864.773
```

9. Torque motor (L = paso hilo, e = eficiencia)

```
In[ ]:= L = 0.01;
e = 0.9;

In[ ]:= T = (F * L) / (2 * Pi * e)
           |número π
Out[ ]:= 1.52925
```